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ORIGINAL ARTICLE

Perceptions of PrEP on Twitter: A Theoretically Guided Content Analysis on the Behavioral Determinants of PrEP Uptake

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HIV pre-exposure prophylaxis (PrEP) is a preventative strategy that involves taking a daily, oral medication to reduce the risk of contracting HIV. Advancements in recent years may have changed the ways in which PrEP is discussed in public discourse, and the use of social media for openly discussing and disseminating PrEP-related promotion messages has been understudied. Thus, this study focuses on understanding the current discourse surrounding PrEP on Twitter. Through a quantitative content analysis, this study examined the messaging, sources, and sentiments and emotions expressed in PrEP-related tweets posted from April 2019 to April 2020, six months before and after the FDA approval of Descovy for PrEP. After the approval of Descovy, there was a significant decrease in tweets relating to barriers for using PrEP; further, informational tweets about PrEP significantly increased. However, a large proportion of tweets still discussed barriers (29.9%), suggesting that addressing barriers to PrEP remains a top priority for promoting PrEP uptake. A lack of tweets discussing individual attitudes, norms, and behavioral control about PrEP may suggest that the social media discourse is not covering the important aspects of behavioral determinants for PrEP adoption. Health communication researchers can focus on developing message strategies that target these theoretical constructs to promote the adoption of PrEP among key target populations.

Keywords: HIV Prevention, Pre-Exposure Prophylaxis, Theory of Planned Behavior, Sentiment Analysis, Message Strategies, Social Media, Twitter.

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Introduction

HIV pre-exposure prophylaxis, or PrEP, is a preventative strategy that involves taking a daily, oral medication to reduce the risk of contracting HIV. Though this method has shown to be effective in preventing HIV acquisition in several clinical trials (Baeten et al., 2012; Choopanya et al., 2013; Grant et al., 2010, 2014; McCormack et al., 2016; Molina et al., 2015, 2017; Thigpen et al., 2012), PrEP uptake is still relatively low in the U.S. (Harris et al., 2019). For example, a recent national survey found that only 7.8% of eligible sexually active gay and bisexual men were taking PrEP in the U.S. (Holloway et al., 2020). Despite some programs that aim to increase access to PrEP, underlying psychological factors, such as the lack of positive beliefs related to taking PrEP (Golub et al., 2019; Meyers et al., 2021), may also influence PrEP uptake. Thus, there is a need to promote PrEP among populations disproportionately affected by HIV through the development of public communication message strategies that address the beliefs and concerns surrounding the medication.

With most US adults on social media (Auxier & Anderson, 2021) and because gay and bisexual men actively use social media (DeHaan et al., 2013; Magee et al., 2012; Patel et al., 2016), examining online discussions surrounding new issues may be useful for assessing people's attitudes and opinions. Utilizing this online approach provides naturalistic data that evades the social desirability biases that often come with traditional survey measures (Babbie, 2016). Further, examining online behavioral data may be advantageous to reach individuals and populations that may otherwise be difficult to contact when using traditional methodologies. Since roughly 42% of young adults ages 18 to 29 use Twitter (Auxier & Anderson, 2021), examining discussions on the social media platform can provide the information necessary to understand the current discourse surrounding PrEP.

To further illustrate the need to examine online discussions surrounding PrEP, the HIV National Strategic Plan for the United States 2021-2025 (U.S. Department of Health and Human Services [HHS], 2021) has prioritized several target goals; the first focusing on preventing new HIV infections. For example, the HHS has proposed strategies to increase the public's awareness of HIV, including developing prevention messages and campaigns that are clear, appropriately tailored, and reflect recent advancements in science, including promoting PrEP for at risk populations. Further, it is suggested that these campaigns are created specifically to be spread through either traditional or social media by expert organizations and influencers. Because of

the growing developments relating to PrEP, as well as the increasing need to develop tailored, up-to-date messaging, further investigation on the current discourse surrounding PrEP is needed.

This study focuses on understanding the discourse surrounding PrEP on Twitter. Thus, we examine discussions involving PrEP six months before and after the Food and Drug Administration's (FDA) approval of Descovy for PrEP on October 3, 2019 (FDA, 2019). Descovy for PrEP is a daily oral pill to reduce one's risk of contracting HIV and is approved for many target populations excluding those who have receptive vaginal sex. With an additional option for PrEP, it expected that the news of the approval of this new prescription brand-name medication would spur social media discussions not only about Descovy, but also HIV prevention and PrEP in general. This created an opportune window to examine people's beliefs toward the medication, as well as the barriers that impede individuals from taking PrEP. This can aid in guiding researchers and public health professionals to promote its use to the populations most at risk of contracting HIV. To examine these beliefs and barriers, the present study utilizes the Theory of Planned Behavior (TPB) as a coding framework for analyzing the tweet messages. In addition, we investigate the sources of these messages to provide a deeper picture in how individuals or organizations may frame, discuss, and promote PrEP differently. Lastly, we examine the sentiment and emotions expressed within the tweets to understand whether users had favorable or unfavorable views toward PrEP. Findings from this study will help guide future interventions and campaigns in developing relevant and theoretically driven message strategies to promote behavioral initiation and uptake of PrEP.

Literature Review

Daily, oral PrEP was first approved by the FDA in 2012 (Centers for Disease Control and Prevention [CDC], 2012), under the brand name Truvada, as a medication to prevent HIV transmission among those who may be at-risk for contracting HIV. In 2014, the CDC later published clinical guidelines for PrEP, offering more specific criteria for prescribing PrEP, as well as advice on how to promote and support those who take the medication (CDC, 2014). While PrEP is a potential game-changer for HIV prevention, there have been several barriers for individuals to initiate the use of PrEP, including its costs and

accessibility (Bauermeister et al., 2013; Grov et al., 2015), as well as the stigma surrounding taking an HIV medication that can be falsely perceived as promoting "promiscuity" (Calabrese & Underhill, 2015).

Several advancements since the initial endorsement of PrEP in 2012 may have changed the ways in which PrEP is perceived by the general public in the past decade. While only 18.1% of eligible PrEP users have ever been prescribed the medication, there has been a steady increase in prescriptions over the past few years (Harris et al., 2019), which suggests that the general public, doctors, and patients' knowledge and awareness of PrEP has shifted. For example, among a probabilistic national sample of gay and bisexual men, PrEP uptake increased from 4.1% in March 2016 to 7.8% in March 2018, while familiarity increased from 59.8% to 92% (Holloway et al., 2020). While many early studies find that awareness is a common barrier to PrEP use (Eaton et al., 2017; Garnett et al., 2018; Marcus et al., 2019), a primary driver for PrEP initiation and uptake may be related to behavioral beliefs (Dai & Calabrese, 2022; Golub et al., 2019; Meyers et al., 2021). As both the general public and at-risk populations become more familiar with the medication, deeper level behavioral beliefs surrounding PrEP need to be understood to further promote behavioral changes for PrEP initiation and adoption for prevention.

Furthermore, the FDA recently approved another medication, Descovy for PrEP (FDA, 2019). This additional medication to prevent HIV may have sparked new concerns surrounding the effectiveness and accessibility of PrEP overall. Thus, it is important to understand the current public discourses on PrEP: what type of messaging is currently most salient, who are posting these messages, and what are the general sentiments and emotions expressed within these discussions. Further, assessing the current barriers to PrEP will allow researchers to focus their resources on the more relevant psychological and environmental constraints. This understanding will provide the information necessary for health professionals to further tailor their intervention and campaign materials and better address the needs and concerns of the population.

Social Media and PrEP

Since the FDA approval of Truvada for PrEP in 2012, there have been several studies examining the online discourse surrounding PrEP (An et al., 2014; Chan et al., 2021; Hannaford et al., 2018; Hill et al., 2018; Kecojevic et al., 2018, 2020; McLaughlin et al., 2016; Schwartz & Grimm, 2017; Walsh-Buhi et

al., 2021). For example, Kecojevic et al. (2018) conducted a content analysis of the 217 most viewed YouTube videos involving PrEP in 2016. The authors found that over 83.4% were promotional videos for PrEP uptake; however, there were a smaller number of videos that discussed potential barriers to PrEP such as access, costs, side effects, or its safety (Kecojevic et al., 2018). Roughly 29.0% of the videos were created by individuals who use PrEP, while the remaining majority were posted by medical, community, or news organizations (Kecojevic et al., 2018). In addition, Walsh et al. (2021) analyzed 250 Instagram posts related to PrEP from April 2019 to April 2020, and found that the majority of the posts provided a definition of what PrEP was, but less than 40% promoted the medication's use or effectiveness. Organizations consisted of 77.2% of all post creators and were more likely to discuss information on how to use PrEP than individual posters; however, individuals were more likely to post about potential side effects (Walsh-Buhi et al., 2021).

Similarly, previous research has examined PrEP on Twitter when the FDA first approved Truvada in 2012 (An et al., 2014; McLaughlin et al., 2016), and when the CDC endorsed PrEP in 2014 (Schwartz & Grimm, 2017). Through an analysis of 774 English tweets posted from November 5 to December 27, 2012, McLaughlin et al. (2016) found that negative affective tone was associated with the spread of PrEP messages, indicating that tweets with a negative tone were more likely to be reposted on Twitter. Within the content of the tweets, about 21.6% specified the target populations for PrEP, 10.2% discussed the effectiveness of the medication, and 6.8% revolved around moral judgements surrounding its use (McLaughlin et al., 2016). Further, in contrast to findings on YouTube and Instagram, about 60.2% of the tweets were made by individuals, while the rest varied between nonprofit, academic, commercial, or news organizations (McLaughlin et al., 2016).

Among the top tweets relating to PrEP, Schwartz and Grimm (2017) analyzed the contents of roughly 1,093 tweets between May 2013 to 2015. The authors found that over half the tweets provided information about PrEP, while 14.7% focuses on barriers to PrEP, 13.6% involved the limitations of PrEP (e.g., does not protect against other STIs), 8.8% were tweets voicing against stigma, and 5.6% included stigmatizing tweets (Schwartz & Grimm, 2017). Barriers against PrEP were relatively matched with arguments challenging them, and included costs, access, side-effects, and adherence (Schwartz & Grimm, 2017). For example, the number of tweets discussing cost

as a barrier to take PrEP were relatively equal to the number of tweets arguing about its affordability, potential creating a sense of uncertainty about PrEP among Twitter users. In addition, 56% of posts were sent by individuals, while the remaining were from organizations (Schwartz & Grimm, 2017).

While these studies provide a broad overview of several social media platform discussions surrounding PrEP, further investigation is needed as scientific advancements and public views toward PrEP may have changed over the years. For example, with the addition of Descovy, there have been no studies to our knowledge that specifically utilized the keyword for examining perceptions of the medication online. In addition, to provide a deeper understanding of users' perceptions of PrEP, further study is necessary to examine large or complete datasets that are representative of the social media platform. Rather than examining a small sample of posts, examining the full dataset of posts or tweets may provide a more accurate depiction of online discourse surrounding a topic. Lastly, further research is necessary to examine individuals' beliefs and barriers to PrEP through a theoretically driven approach. In this way, researchers and public health professionals can focus on the specific theoretical constructs that predict behavior change, and further develop messages strategies based on the concerns and needs mentioned by social media users.

Theory of Planned Behavior

The Theory of Planned Behavior (TPB) is a behavior change model that provides a useful framework for examining individual's perceptions of PrEP (Fishbein & Ajzen, 2011). TPB has been widely used in studies as a framework to examine PrEP-related behaviors (Roth et al., 2019; Tran et al., 2021; Wang et al., 2020). TPB posits that one's attitudes, perceived norms, and perceived behavioral control may influence one's intention to perform a behavior, which in turn, will lead to the behavior (Fishbein & Ajzen, 2011). An attitude can be defined as one's positive or negative evaluation of a behavior (Fishbein & Ajzen, 2011). For example, if an individual believes that PrEP is a good medication and is useful, the individual will have a positive attitude toward PrEP. Perceived norms can be defined as concerning both injunctive or descriptive norms: injunctive norms refer to whether one believes others would expect them to perform a behavior, and descriptive norms refer the perception of whether others are performing a behavior (Fishbein & Ajzen, 2011). For example, if someone believes others would expect them to take

PrEP, that person would have a high perceived injunctive norm; whereas, if someone believes that most of their friends are taking PrEP, then that person would have a high perceived descriptive norm. Perceived behavioral control refers to whether one believes they have the ability or confidence to perform a behavior (Fishbein & Ajzen, 2011). For example, if an individual does not believe they are able to take PrEP consistently every day, they would have low perceived behavioral control. Intention refers to one's inclination to perform an behavior (Fishbein & Ajzen, 2011), such as one's believe that they are likely to start taking PrEP. Intention is the most direct belief toward actual behavior.

In addition to beliefs, there are other factors that may influence behavior. First, knowledge or awareness may play a direct role in influencing one's beliefs and behaviors (Fishbein & Ajzen, 2011). For example, without the knowledge of HIV or PrEP, one may not know the risks of HIV or HIV-preventative options. Lastly, and one of the most important, environmental constraints may lower one's behavioral intention and even impede one from performing a behavior. As previously mentioned, barriers such as costs, access, and stigma may all prevent individuals from taking PrEP (Bauermeister et al., 2013; Calabrese & Underhill, 2015; Grov et al., 2015).

Because of the comprehensive nature of this theoretical model for behavior prediction, it provides an excellent framework to analyze the beliefs surrounding online posts relating to PrEP. For this study, we focus on six distinct theoretically derived constructs: attitudes, norms, behavioral control, intention/behavior, knowledge/awareness, and actual barriers. To examine actual barriers, we also focused on four environmental constraints: cost, access, issues with the pharmaceutical company, and stigma, based on previous research (McLaughlin et al., 2016; Schwartz & Grimm, 2017). This allows us to pinpoint the exact external barriers that may impact individuals' PrEP uptake decisions. Through analyzing these different dimensions of the behavior model, researchers can pinpoint which constructs are necessary to focus on in future interventions. Thus, the following research questions are proposed:

RQ1: Under the theoretical lens of the TPB, what behavioral predictors for PrEP are discussed on Twitter for HIV prevention among tweets posted six months before and after the approval of Descovy?

RQ2: Which barriers are most discussed on Twitter among tweets posted six months before and after the approval of Descovy?

Source Characteristics

In addition to beliefs expressed online, another primary component of Twitter posts is the source. Expert sources may be influential in disseminating information, and source credibility has shown to be a persuasive cue compared to sources with low credibility (Pornpitakpan, 2004). For example, Zhang et al. (2019) experimentally tested whether organizational versus personal online posts relating to cervical cancer were more likely to be shared, and found that organizational posts were more likely to be shared. However, individuals can give personal narratives with compelling stories that may impact one's behaviors (Kecojevic et al., 2020; Surian et al., 2016; Zhang et al., 2019).

Further, while source type may influence one's behaviors and beliefs associated with those behaviors, providing a full picture of the current discourse surrounding PrEP will allow researchers and practitioners to have a baseline understanding of which actors are involved in online media. Previous research has found that over half of the tweets related to PrEP were from individuals rather than organizations (McLaughlin et al., 2016; Schwartz & Grimm, 2017). This implies that individuals may be sharing more personal stories related to PrEP in comparison to health organizations, suggesting a shift in public awareness of PrEP. As PrEP becomes more known by the public, it is important to understand whether the source types have shifted over time. The following research question is proposed:

> RQ3: Under the theoretical lens of the TPB, which source type (organization vs. individual) is more likely to discuss the behavioral predictors of PrEP among tweets posted six months before and after the approval of Descovy?

Sentiments and Emotions

Further, we are interested in examining the sentiment of Twitter discussions surrounding PrEP within the year period. Previous work has examined sentiments of online posts for various health and science topics, including vaping (Martinez et al., 2018), HPV vaccination (Kearney et al., 2019; Massey et al., 2020; Zhang et al., 2021), and gene editing (Calabrese et al., 2019, 2020). To our knowledge, research examining PrEP discussions on Twitter have yet to specifically measure both the expressed sentiments and emotions.

Messages that express certain sentiments or emotions may have an impact on individuals' beliefs toward a health topic; a negative sentiment toward PrEP

may deter others from becoming willing to start taking the medication. For example, one study found that state-level exposure to HPV vaccine information on Twitter was both positively and negatively associated with vaccine coverage depending on the valence of the topic discussed (Dunn et al., 2017). It is also important to understand how Twitter users feel about PrEP, and whether future work is needed to further promote the positive aspects of the preventative medication. Thus, we propose the research question:

> RQ4: What are the sentiments and emotions expressed among PrEP-related tweets posted six months before and after the approval of Descovy?

Approval of Descovy

Lastly, this study examines the messaging, sources, and sentiments and emotions of tweets related to PrEP before and after the approval of Descovy. The addition of another medication for PrEP, though only approved for certain populations, may change the ways in which PrEP is discussed, who are discussing PrEP, and the general sentiments and emotions expressed. Previous research has found that the contents and sentiments of discussions surrounding health and science topics may change after news or large events (Calabrese et al., 2020; Guidry et al., 2020). Thus, we propose the final research question:

> RQ5: How do the messaging, sources, and sentiments and emotions change after the approval of Descovy?

Methods

Data Collection

We conducted a quantitative content analysis to examine tweets discussing PrEP from April 2019 to April 2020, six months before and after the approval of Descovy in October 2019 (FDA, 2019). Using the R package *twint* (Zacharias, 2020), a comprehensive list of English-language tweets was collected with the search terms "pre-exposure prophylaxis," "Truvada," and "Descovy," as well as the hashtag "#PrEP." These hashtags were utilized based off of search terms used in previous research (McLaughlin et al., 2016; Schwartz & Grimm, 2017), as well as a preliminary search on Twitter. The hashtag "PrEP" was chosen over the keyword "PrEP," because the vast majority of search results for the

keyword were unrelated (e.g., meal prep). Only original tweets were collected (retweets were excluded from analysis). After the removal of duplicates, there were 16,138 tweets that resulted. The dataset was collected in June 2020.

Coding Framework and Analysis

A content analysis involves the systematic analysis of message characteristics within a given text (Babbie, 2016). For our content analysis, we conducted systematic random sampling where every 16th tweet was selected; this allowed each tweet to have an equal chance in being selected, ensuring randomization. The resulting sample of 1,008 tweets was manually coded based on our codebook. A coding framework for the message contents was developed based on TPB (Fishbein & Ajzen, 2011); see Table 1 for categories, their definitions, their reliability, and example tweets. The coding categories of theoretical constructs include attitudes, norms, perceived behavioral control, intention/behavior, information/knowledge, and barriers. Among barriers, we coded sub-categories including costs, access, issues with the pharmaceutical companies, and stigma. These barriers were chosen based off previous research (McLaughlin et al., 2016; Schwartz & Grimm, 2017). For each category, contents were coded as either present "1" or absent "0."

Source type was coded as either an individual or organization. For example, LGBT News and CDC were coded as organizations, while accounts with full names (including public figures) were coded as individuals. Previous work has separately categorized different organizations, such as news media and community based organizations (McLaughlin et al., 2016; Schwartz & Grimm, 2017; Walsh-Buhi et al., 2021); however, this study examines organizations as one category to make meaningful comparisons with individuals for analyses.

After pretesting and training, two coders independently coded about 20% of the randomly selected sample (N=209) and achieved a satisfactory inter-coder reliability (Cohen's kappa ranging from .76 to 1.0; see Table 1 for individual reliabilities) (McHugh, 2012). The coders then coded about 400 remaining tweets each. Chi-squared tests were conducted to examine differences before and after the approval of Descovy, as well as differences by source type. For content categories with expected values frequencies of less than five, we conducted Fisher's exact tests.

Table 1. Coding Categories and Examples

Category	Definition	Reliability	Example
TPB Construct			
Attitude	Positive or negative evaluation of PrEP	.81	Stupid ass Truvada is causing these really annoying headaches at night.
Norm	Perception of what others expect people take PrEP or what others are taking PrEP.	1.0	Many of us have already been switched, @GileadSciences CEO reported that already 25% of PrEP pts are on Descovy rather than Truvada. This is consistent with what @PrEP4AllNow has found, with Descovy Rxs surging more than 70% since Descovy was FDA approved back in October.
PBC	Perceptions of the level of control or capabilities for taking PrEP.	1.0	That's where you make empowered decisions through your own discernment. Truvada ain't the only PrEP option.
Intention/ Behavior	Likelihood of taking PrEP or are taking PrEP.	.76	Day 2 on Descovy i hope my white blood cells are chillin
Information/ Knowledge	Information about PrEP.	.91	What is PrEP? #PrEP stands for pre-exposure prophylaxis. This involves taking medicine that can give you protection against HIV if you have unprotected sex with someone who is HIV-positive.
Actual barriers	Barriers or facilitators for taking PrEP.	-	[Examples provided below.]
Cost	Issues with cost of PrEP.	.92	Rep. Alexandria Ocasio-Cortez (D-N.Y.) confronted a CEO Thursday for pricing the anti-HIV drug Truvada, aka, PrEP, at $8 in Australia but

			over $1,500 in the U.S.
Access	Issues related to PrEP access not directly related to costs.	.80	Gilead will donate HIV prevention Truvada drug to 200k Americans
Issues with Pharmaceutical Company	Issues with the pharmaceutical company including mistrust or patents.	.87	The Pharma Industry is a huge Mafia Cartel. They don't necessarily create cures, they only create more customers!
Stigma	Issues related to stigmatizing beliefs.	.83	Does anyone else besides me think the Truvada commercials not are only nauseating, but sends a wrong message to our youth. Wtf..has this country become.. liberals are destroying the future and indoctrinating the young.

Notes: Content categories were coded as either present "1" or absent "0" within each tweet. Reliabilities were assessed using Cohen's kappa.

Linguistic Analysis

We ran Linguistic Inquiry and Word Count (LIWC) to examine the sentiment of all 16,138 tweets; LIWC is a validated dictionary-based analysis software that provides the emotions and sentiments of words in a message (Pennebaker et al., 2015; Tausczik & Pennebaker, 2010). To obtain the positive and negative emotions expressed in the tweets, LIWC outputs the percentage of positive and negative emotion words within each tweet; this allows us to control for word count. Several studies have utilized LIWC to examine the expressed emotions and sentiments of online health discussions (Rains et al., 2021; Zhang et al., 2021). For example, Zhang and colleagues (2021) examined the emotion and sentiments expressed by public Facebook groups and pages relating to the HPV vaccine. To examine differences between emotional language used before and after the approval of Descovy, we conducted Welch's t tests, which account for unequal variances and sample sizes, utilizing the full 16k dataset.

Lastly, subsequent analyses were conducted to ensure the findings were robust. Mann Whitney U tests were conducted to provide stricter criteria for

all comparisons. Further, we also used Botometer (Sayyadiharikandeh et al., 2020) to classify tweets that were likely created by bots and re-ran all analyses. Except for one finding, there were no significant differences between the "no bot" and regular sample results. The results can be found in the Appendix. All analyses were conducted in SPSS Version 27.

Results

Figure 1 depicts the number of tweets that mentioned PrEP over time by month. There were about 8,402 tweets before the approval of Descovy and 8,097 tweets posted after the approval. Aside from the sentiment and emotion analyses, the following results are based on the systematic random sample of coded tweets.

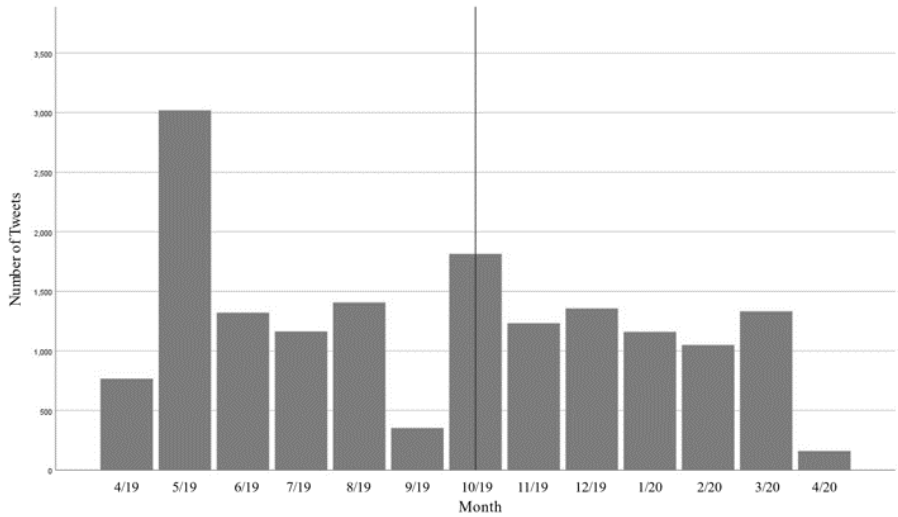


Figure 1. Number of Tweets Discussing PrEP over Time

To address RQ1 and RQ2, we examined the behavioral predictors and specific barriers discussed about PrEP. Overall, among the sample of 1,008 coded tweets, most were related to barriers (38.2%). Among the barriers (N=385), 36.6% were related to costs, 36.4% focused on accessibility, 21.3% delved into issues related to the pharmaceutical companies, and 5.7% related to stigma and stigmatizing tweets.

Roughly 25.8% of tweets discussed information or knowledge about PrEP,

while 7.9% expressed PrEP uptake behaviors or intentions. Lastly, only 6.2% expressed attitudes toward PrEP, 1.2% discussed norms supporting PrEP, and 1.7% discussed behavioral control with taking PrEP. For source type (RQ3), most of the tweets were composed by individuals (74.9%), while organizations consisted of about only one fourth of the tweets.

Regarding the sentiments expressed in the full sample of tweets (RQ4), the average percentage of positive emotion expressed within each tweet was 2.01 (SD = 3.06), while the average negative emotion expressed in each tweet was 1.34 (SD = 2.66). For the three discrete emotions, the average percentage of emotion expressed in each tweet was .33 (SD = 1.0) for anxiety, .34 (SD = 1.44) for anger, and .21 (SD = .96) for sadness. This means that on average about 2% of each tweet involved positive words, 1.3% involved negative words, and less than 1% involved anxiety, anger, and sadness words.

Comparisons by Time Period

Our final research question (RQ5) focused on the differences in messaging, sources, and sentiment before and after the approval of Descovy.

TPB Constructs and Source Type

The results of the content analyses comparing differences between time periods can be found in Table 2. We found that there was a significant decrease in barriers discussed after the FDA approval of Descovy, $\chi^2(1, N=1008) = 26.57$, $p < .001$. Furthermore, the number of informational posts increased after the FDA approval, $\chi^2(1, N=1008) = 9.10$, $p = .003$. There were no significant differences in tweets for attitudes, norms, behavioral control, or intention/behavior.

Table 2. Characteristics of PrEP Tweets by FDA Approval of Descovy Time Period (N = 1008)

Variable	Before FDA Approval N (%)	After FDA Approval N (%)	p-value
TPB Construct			
Attitude	28 (5.3%)	34 (7.1%)	.247
Perceived Norm	9 (1.7%)	3 (.6%)	.113
Perceived behaviora control	6 (1.1%)	11 (2.3%)	.157
Intention/Behavior	35 (6.6%)	45 (9.4%)	.111
Information/ Knowledge	115 (21.8%)	145 (30.3%)	.003
Actual barriers	241 (45.7%)	144 (29.9%)	<.001
Other	93 (17.6%)	99 (20.6%)	.236
Source			
Individual	392 (74.4%)	363 (75.5%)	.692
Organization	135 (25.6%)	118 (24.5%)	

Notes: P-values were calculated from χ^2 tests; all cells had expected values over 5.

Among the specific barriers (See Table 3), there was a significant decrease in posts regarding access to the medication $\chi^2(1, N=385) = 5.15$, $p = .023$. There were no significant differences in cost, issues with the pharmaceutical companies, or stigma.

In addition, there were no differences between individual and organizational posts after the approval of Descovy.

Table 3. Types of Actual Barriers by FDA Approval of Descovy Time Period (N = 385)

Variable	Before FDA Approval N (%)	After FDA Approval N (%)	p-value
Cost	85 (35.3%)	56 (38.9%)	.476
Access	98 (40.7%)	42 (29.2%)	.023
Issues with Pharmaceutical Company	48 (19.9%)	34 (23.6%)	.392
Stigma	10 (4.1%)	12 (8.3%)	.087

Notes: P-values were calculated from χ^2 tests; all cells had expected values over 5.

Sentiments and Emotions

Compared to before the FDA approval of Descovy (M = 1.72, SD = 2.91), the percentage of positive emotion significantly increased (M = 2.11, SD = 3.27), t(15944.10) = -7.96, p < .001. There was no significant difference in expressed negative emotion. For discrete emotions, the percentage of anger expressed in tweets decreased from before (M = .43, SD = 1.70) to after the FDA approval (M= .33, SD = 1.44), t(15686.81) = 4.03, p < .001. There were no significant differences in anxiety or sadness.

Comparisons by Source Type

Lastly, to delve deeper into our research questions, we also examined differences in behavioral predictors and sentiments and emotions by source type.

TPB Constructs

Table 4 compares the TPB constructs by source type. Among organizational tweets, information (44.7%) and barriers (40.3%) were most discussed, while less than 2% of the tweets discussed attitude, norms, behavioral control, or intention/behavior. Similarly, most individual tweets discussed barriers (37.5%) and information (19.3%), while roughly 10.1% of the tweets discussed intention/behavior, 8.1% discussed attitude, 2.1% discussed behavioral control, and 1.2% discussed norms.

Individuals discussed their attitudes toward PrEP ($\chi^2(1, N=1008) = 19.38$, $p < .001$) and PrEP uptake behaviors or intentions ($\chi^2(1, N=1008) = 18.67$, $p < .001$) significantly more than organizations. However, organizations provided a significantly higher number of informational tweets ($\chi^2(1, N=1008) = 62.84$, $p < .001$) than individuals. There were no differences between the two source types for norms and behavioral control, as well as barriers.

When examining specific barriers, individuals were more likely to post about the costs of PrEP, $\chi^2(1, N=385) = 6.16$, $p = .013$, and stigma associated with PrEP, $\chi^2(1, N=385) = 5.77$, $p = .016$, compared to organizations. However, organizations were more likely to post about issues with the accessibility of PrEP, $\chi^2(1, N=385) = 12.81$, $p < .001$. There were no differences in source type with regard to issues with pharmaceutical companies.

Table 4. Characteristics of PrEP Tweets by Source (N = 1008)

Variable	Organization	Individual	p-value
TPB Construct			
Attitude	1 (.4%)	61 (8.1%)	<.001
Perceived norm	3 (1.2%)	9 (1.2%)	1.0 [a]
Perceived behavioral control	1 (.4%)	16 (2.1%)	.088 [a]
Intention/behavior	4 (1.6%)	76 (10.1%)	<.001
Information/ knowledge	113 (44.7%)	146 (19.3%)	<.001
Actual barriers	102 (40.3%)	283 (37.5%)	.422
Other	29 (11.5%)	163 (21.6%)	<.001

Notes: P-values were calculated from χ^2 tests; all cells had expected values over 5.

[a] P-values were calculated using Fisher's exact tests.

Sentiments and Emotions

Individual posters expressed a higher percentage of positive emotion words (M = 2.22, SD = 3.29) compared to organizations (M = 1.48, SD = 2.18), t(654.09)= -3.95, p < .001. However, there were no significant differences in negative emotion by source type.

Organizations expressed significantly more anxiety (M = .45, SD = 1.17) than individual posters (M = .29, SD = .94), t(366.92) = 1.97, p = .049. There were no significant differences between individuals and organizations for anger or sadness.

Discussion

This study examined the content and sources of tweets related to PrEP through a systematic random sample of tweets, as well as the sentiments expressed within all tweets that mentioned PrEP between April 2019 to April 2020. By examining these three main features of the social media posts, researchers can identify key socio-behavioral factors that may influence behavior change, specifically relating to PrEP uptake. Further, there were

several differences before and after the approval of Descovy.

Overall, about 38.2% of tweets discussed related to barriers to PrEP, among which cost and access were most discussed. Our results indicate after the approval of Descovy, there was a significant decrease in the discussions surrounding barriers to PrEP; specifically, there was a decrease in posts regarding issues with access. Despite this reduction, a large proportion of posts still discussed barriers after the FDA approval. Barriers directly hinder people's ability to start taking PrEP and adhere to the medication. This may be a factor in why PrEP uptake remains relatively low.

Most of the barriers discussed refer to costs, access, and issues related to the pharmaceutical companies. There are several federal and state assistance programs that provide individuals with the medication at a significantly reduced cost. For example, the "Ready, Set, PrEP," program will provide the medication at no costs for those who are uninsured. However, there are different eligibility criteria and several steps to enroll in a program, and many of these programs do not also cover the required laboratory tests and clinical visits. Public health practitioners may focus on providing information to the public on how to afford PrEP and access these resources. Further, efforts may also focus on reducing cost barriers not directly related to the medication, including lab and clinic visits. The decrease in tweets relating to access issues serves as a positive sign, especially with generic versions of PrEP now available (Highleyman, 2020, 2021), but future work must be done to reduce barriers for those who need the medication the most.

The percentage of positive emotion words in PrEP discussions significantly increased after the approval of Descovy, indicating that continuing to increase access to the medication may be viewed favorably by Twitter users. In addition, anger decreased after the approval of Descovy, potentially indicating that individuals may feel some relief after having an additional option to access PrEP. Because emotion may play a factor for behavior change, a more favorable view toward PrEP may help in improving PrEP uptake. Similarly, the reduction of negative emotions may also aid in the adoption of PrEP. The changes after the approval of another medication serves as an example that increasing access to a medication can really have a large positive impact, and further illustrates the need for policies to enable individuals to obtain PrEP and eliminate barriers.

Compared to organizations, individuals were more likely to discuss their attitudes and intentions surrounding PrEP and less likely to post

informational tweets. While this makes intuitive sense, organizations may need to start developing messages that are tailored toward people's beliefs to promote PrEP adoption. Rather than only providing information, targeting beliefs about PrEP will have a larger impact for improving uptake. Thus, with TPB as the guiding framework for the study, having messages relating to attitudes, norms, or behavioral control may contribute to behavior change. For example, messages that indicate that most gay and bisexual men have favorable attitudes toward taking PrEP may be helpful in promoting the medication to the public. Developing messages that target theoretical constructs and are relevant to target populations may help increase PrEP adoption rates.

Further, individuals were more likely to express positive emotion compared to organizations, while organizations were more likely to express anxiety. This may partially be due to the language used when organizations are communicating about the risks of contracting HIV. However, organizations may need to pay particular attention on how they are constructing their messages to ensure that they are showing PrEP in a favorable light, rather than a neutral or negative tone.

While the discussion of barriers was no different between individuals and organizations, among specific barriers, individuals were more likely to discuss the costs and stigma related to PrEP, while organizations were more likely to discuss access to PrEP. Organizations should continue their efforts in addressing the costs of the medication, as well as addressing the stigma associated with HIV-preventive behaviors. Promoting PrEP should be one part of an organization's HIV prevention toolkit; comprehensive efforts with messaging that includes condom use, HIV testing, PrEP uptake, and HIV stigma would be very beneficial.

Lastly, there was a lack of tweets that involved attitudes, norms, and behavioral control. This sparks a need for future work in addressing these beliefs in relation to PrEP uptake. For example, interventions that harness social influence may be particularly helpful for promoting PrEP. Attitudes, norms, and behavioral control are three primary socio-behavioral factors that influence behavior change, and future research can focus on developing message strategies that address these beliefs.

Limitations and Future Directions

There are several limitations to this study. First, this study examined perceptions of PrEP through only one social media platform during one time point. Twitter is not a representative sample of the general public (Wojcik & Hughes, 2019), nor is it representative of the social media landscape. Our study provides an example of using a theoretical framework to content analyze the social media posts. Further research is necessary to understand perceptions of PrEP in other online contexts, such as anonymous support forums and groups. We also did not content analyze the images and videos that may be attached to tweets; however, it is believed that the initial focus of the social media platform is to convey short text-based messages to audiences and an analysis of the texts in tweets may be ideal. Previous work has examined the contents related to PrEP on image and video-based platforms such as Instagram and YouTube (Kecojevic et al., 2018, 2020; Walsh-Buhi et al., 2021). In addition, while the study focused on the differences before and after the approval of Descovy, future work can examine perceptions of PrEP over time as new advancements in HIV prevention develop. Lastly, the study utilized LIWC (Pennebaker et al., 2015), a dictionary-based tool to assess expressed emotions and sentiments. While the software has been validated in previous studies, recently developed new machine learning tools may provide a more sophisticated output for sentiments expressed in messages. Despite these limitations, findings from this study provide a detailed analysis of the perceptions, sentiments, and emotions of Twitter discussions surrounding PrEP, as well as the sources that disseminate these tweets, which will help inform future interventions for promoting PrEP.

Conclusion

This study examined the messaging, sources, and sentiments and emotions of PrEP discussions on Twitter over a one-year time period. We found that the number of tweets discussing barriers decreased after the approval of Descovy, specifically relating to the accessibility of the medication. However, barriers still dominated the discussion of PrEP overall. With recent news regarding the availability of generic PrEP, health practitioners should continue focusing on reducing barriers to PrEP.

Further, the number of tweets related to attitudes, norms, and behavioral control were limited on the platform, especially from organizational sources. Health communication researchers can focus on developing message strategies that target these constructs to promote the adoption of PrEP among key target populations. By examining the discussions through the lens of TPB, one can pinpoint specific beliefs related to behavior change, which may better inform future health interventions.

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Declaration of interest statement

The authors report there are no competing interests to declare.

References

An, Z., McLaughlin, M., Hou, J., Nam, Y., Hu, C.-W., Park, M., & Meng, J. (2014). Social network representation and dissemination of pre-exposure prophylaxis (PrEP): A semantic network analysis of HIV prevention drug on Twitter. In G. Meiselwitz (Ed.), *Social Computing and Social Media* (pp. 160–169). Springer International Publishing. https://doi.org/10.1007/978-3-319-07632-4_15

Auxier, B., & Anderson, M. (2021). Social media use in 2021. *Pew Research Center*. Retrieved from https://www.pewresearch.org/internet/2021/04/07/social-media-use-in-2021/

Babbie, E. R. (2016). *The basics of social research* (7th edition). Cengage Learning.

Baeten, J. M., Donnell, D., Ndase, P., Mugo, N. R., Campbell, J. D., Wangisi, J., Tappero, J. W., Bukusi, E. A., Cohen, C. R., Katabira, E., Ronald, A., Tumwesigye, E., Were, E., Fife, K. H., Kiarie, J., Farquhar, C., John-Stewart, G., Kakia, A., Odoyo, J., … Celum, C. (2012). Antiretroviral prophylaxis for HIV prevention in heterosexual men and women. *New England Journal of Medicine, 367*(5), 399–410. https://doi.org/10.1056/NEJMoa1108524

Bauermeister, J. A., Meanley, S., Pingel, E., Soler, J. H., & Harper, G. W. (2013). PrEP awareness and perceived barriers among single young men who have sex with men in the United States. *Current HIV Research, 11*(7), 520–527.

Calabrese, C., Anderton, B. N., & Barnett, G. A. (2019). Online representations of "genome editing" uncover opportunities for encouraging engagement: A semantic network analysis. *Science Communication, 41*(2), 222–242. https://doi.org/10.1177/1075547018824709

Calabrese, C., Ding, J., Millam, B., & Barnett, G. A. (2020). The uproar over gene-edited babies: A semantic network analysis of CRISPR on Twitter. *Environmental*

Communication, *14*(7), 954–970. https://doi.org/10.1080/17524032.2019.1699135

Calabrese, S. K., & Underhill, K. (2015). How stigma surrounding the use of HIV preexposure prophylaxis undermines prevention and pleasure: A call to destigmatize "truvada whores." *American Journal of Public Health, 105*(10), 1960–1964. https://doi.org/10.2105/AJPH.2015.302816

Centers for Disease Control and Prevention. (2012). CDC statement on FDA approval of drug for HIV prevention. Retrieved from https://www.cdc.gov/nchhstp/newsroom/2012/fda-approvesdrugstatement.html

Centers for Disease Control and Prevention. (2014). New guidelines recommend daily HIV prevention pill for those at substantial risk. Retrieved from https://www.cdc.gov/nchhstp/newsroom/2014/prep-guidelines-press-release.html

Chan, M. S., Morales, A., Zlotorzynska, M., Sullivan, P., Sanchez, T., Zhai, C., & Albarracín, D. (2021). Estimating the influence of Twitter on pre-exposure prophylaxis use and HIV testing as a function of rates of men who have sex with men in the United States. *AIDS, 35*, S101. https://doi.org/10.1097/QAD.0000000000002838

Choopanya, K., Martin, M., Suntharasamai, P., Sangkum, U., Mock, P. A., Leethochawalit, M., Chiamwongpaet, S., Kitisin, P., Natrujirote, P., Kittimunkong, S., Chuachoowong, R., Gvetadze, R. J., McNicholl, J. M., Paxton, L. A., Curlin, M. E., Hendrix, C. W., & Vanichseni, S. (2013). Antiretroviral prophylaxis for HIV infection in injecting drug users in Bangkok, Thailand (the Bangkok Tenofovir Study): A randomised, double-blind, placebo-controlled phase 3 trial. *The*

Lancet, 381(9883), 2083–2090. https://doi.org/10.1016/S0140-6736(13)61127-7

Dai, M., & Calabrese, C. (2022). Socio-behavioral factors related to PrEP non-adherence among gay male PrEP users living in California and New York: A behavioral theory informed approach. *Journal of Behavioral Medicine*. https://doi.org/10.1007/s10865-021-00275-1

DeHaan, S., Kuper, L. E., Magee, J. C., Bigelow, L., & Mustanski, B. S. (2013). The interplay between online and offline explorations of identity, relationships, and sex: A mixed-methods study with LGBT youth. *Journal of Sex Research*. https://doi.org/10.1080/00224499.2012.661489

Dunn, A. G., Surian, D., Leask, J., Dey, A., Mandl, K. D., & Coiera, E. (2017). Mapping information exposure on social media to explain differences in HPV vaccine coverage in the United States. *Vaccine, 35*(23), 3033–3040. https://doi.org/10.1016/j.vaccine.2017.04.060

Eaton, L. A., Matthews, D. D., Driffin, D. D., Bukowski, L., Wilson, P. A., Stall, R. D., & The POWER Study Team. (2017). A multi-US city assessment of awareness and uptake of pre-exposure prophylaxis (PrEP) for HIV prevention among Black men and transgender women who have sex with men. *Prevention Science, 18*(5), 505–516. https://doi.org/10.1007/s11121-017-0756-6

Fishbein, M., & Ajzen, I. (2011). Predicting and changing behavior: The reasoned action approach. *Psychology Press*. https://doi.org/10.4324/9780203838020

Garnett, M., Hirsch-Moverman, Y., Franks, J., Hayes-Larson, E., El-Sadr, W. M., & Mannheimer, S. (2018). Limited awareness of pre-exposure prophylaxis among black men who have sex with men and transgender women in New York city. *AIDS Care, 30*(1), 9–17. https://doi.org/10.1080/09540121.2017.1363364

Golub, S. A., Fikslin, R. A., Goldberg, M. H., Peña, S. M., & Radix, A. (2019). Predictors of

PrEP uptake among patients with equivalent access. *AIDS and Behavior, 23*(7), 1917–1924. https://doi.org/10.1007/s10461-018-2376-y

Grant, R. M., Anderson, P. L., McMahan, V., Liu, A., Amico, K. R., Mehrotra, M., Hosek, S., Mosquera, C., Casapia, M., Montoya, O., Buchbinder, S., Veloso, V. G., Mayer, K., Chariyalertsak, S., Bekker, L.-G., Kallas, E. G., Schechter, M., Guanira, J., Bushman, L., … Glidden, D. V. (2014). Uptake of pre-exposure prophylaxis, sexual practices, and HIV incidence in men and transgender women who have sex with men: A cohort study. *The Lancet Infectious Diseases, 14*(9), 820–829. https://doi.org/10.1016/S1473-3099(14)70847-3

Grant, R. M., Lama, J. R., Anderson, P. L., McMahan, V., Liu, A. Y., Vargas, L., Goicochea, P., Casapía, M., Guanira-Carranza, J. V., Ramirez-Cardich, M. E., Montoya-Herrera, O., Fernández, T., Veloso, V. G., Buchbinder, S. P., Chariyalertsak, S., Schechter, M., Bekker, L.-G., Mayer, K. H., Kallás, E. G., … Glidden, D. V. (2010). Preexposure chemoprophylaxis for HIV prevention in men who have sex with men. *New England Journal of Medicine, 363*(27), 2587–2599. https://doi.org/10.1056/NEJMoa1011205

Grov, C., Whitfield, T. H. F., Rendina, H. J., Ventuneac, A., & Parsons, J. T. (2015). Willingness to take PrEP and potential for risk compensation among highly sexually active gay and bisexual men. *AIDS and Behavior, 19*(12), 2234–2244. https://doi.org/10.1007/s10461-015-1030-1

Guidry, J. P. D., Vraga, E. K., Laestadius, L. I., Miller, C. A., Occa, A., Nan, X., Ming, H. M., Qin, Y., Fuemmeler, B. F., & Carlyle, K. E. (2020). HPV vaccine searches on Pinterest: Before and after Pinterest's actions to moderate content. *American Journal of Public Health, 110*(S3), S305–S311. https://doi.org/10.2105/AJPH.2020.305827

Hannaford, A., Lipshie-Williams, M., Starrels, J. L., Arnsten, J. H., Rizzuto, J., Cohen, P., Jacobs, D., & Patel, V. V. (2018). The use of online posts to identify barriers to and facilitators of HIV pre-exposure prophylaxis (PrEP) among men who have sex with men: A comparison to a systematic review of the peer-reviewed literature. *AIDS and Behavior, 22*(4), 1080–1095. https://doi.org/10.1007/s10461-017-2011-3

Harris, N. S., Johnson, A. S., Huang, Y.-L. A., Kern, D., Fulton, P., Smith, D. K., Valleroy, L. A., & Hall, H. I. (2019). Vital signs: Status of human immunodeficiency virus testing, viral suppression, and HIV preexposure prophylaxis — United States, 2013–2018. *Morbidity and Mortality Weekly Report, 68*. https://doi.org/10.15585/mmwr.mm6848e1

Highleyman, L. (2020, October 2). First generic Truvada now available in the United States. *POZ*. https://www.poz.com/article/first-generic-truvada-now-available-united-states

Highleyman, L. (2021, May 21). Cheaper generic PrEP now available in the United States. *POZ*. https://www.poz.com/article/cheaper-generic-prep-now-available

Hill, B. S., Patel, V. V., Haughton, L. J., & Blackstock, O. J. (2018). Leveraging social media to explore Black women's perspectives on HIV pre-exposure prophylaxis. *The Journal of the Association of Nurses in AIDS Care : JANAC, 29*(1), 107–114. https://doi.org/10.1016/j.jana.2017.06.010

Holloway, I. W., Krueger, E. A., Meyer, I. H., Lightfoot, M., Frost, D. M., & Hammack, P. L. (2020). Longitudinal trends in PrEP familiarity, attitudes, use and discontinuation among a national probability sample of gay and bisexual men, 2016–2018. *PLOS ONE, 15*(12), e0244448. https://doi.org/10.1371/journal.pone.0244448

Kearney, M. D., Selvan, P., Hauer, M. K., Leader, A. E., & Massey, P. M. (2019). Characterizing HPV vaccine sentiments and content on Instagram. *Health Education & Behavior, 46*(2_suppl), 37S-48S. https://doi.org/10.1177/1090198119859412

Kecojevic, A., Basch, C., Basch, C., & Kernan, W. (2018). Pre-exposure prophylaxis YouTube videos: Content evaluation. *JMIR Public Health and Surveillance, 4*(1), e7733. https://doi.org/10.2196/publichealth.7733

Kecojevic, A., Meleo-Erwin, Z. C., Basch, C. H., & Hammouda, M. (2020). A thematic analysis of pre-exposure prophylaxis (PrEP) YouTube videos. *Journal of Homosexuality, 0*(0), 1–22. https://doi.org/10.1080/00918369.2020.1712142

Magee, J. C., Bigelow, L., DeHaan, S., & Mustanski, B. S. (2012). Sexual health information seeking online: A mixed-methods study among lesbian, gay, bisexual, and transgender young people. *Health Education & Behavior, 39*(3), 276–289. https://doi.org/10.1177/1090198111401384

Marcus, J. L., Hurley, L. B., Dentoni-Lasofsky, D., Ellis, C. G., Silverberg, M. J., Slome, S., Snowden, J. M., & Volk, J. E. (2019). Barriers to preexposure prophylaxis use among individuals with recently acquired HIV infection in Northern California. *AIDS Care, 31*(5), 536–544. https://doi.org/10.1080/09540121.2018.1533238

Martinez, L. S., Hughes, S., Walsh-Buhi, E. R., & Tsou, M.-H. (2018). "Okay, we get it. You vape": An analysis of geocoded content, context, and sentiment regarding e-cigarettes on Twitter. *Journal of Health Communication, 23*(6), 550–562. https://doi.org/10.1080/10810730.2018.1493057

Massey, P. M., Kearney, M. D., Hauer, M. K., Selvan, P., Koku, E., & Leader, A. E. (2020). Dimensions of misinformation about the HPV vaccine on Instagram: Content and network analysis of social media characteristics. *Journal of Medical Internet*

Research, 22(12), e21451. https://doi.org/10.2196/21451

McCormack, S., Dunn, D. T., Desai, M., Dolling, D. I., Gafos, M., Gilson, R., Sullivan, A. K., Clarke, A., Reeves, I., Schembri, G., Mackie, N., Bowman, C., Lacey, C. J., Apea, V., Brady, M., Fox, J., Taylor, S., Antonucci, S., Khoo, S. H., … Gill, O. N. (2016). Pre-exposure prophylaxis to prevent the acquisition of HIV-1 infection (PROUD): Effectiveness results from the pilot phase of a pragmatic open-label randomised trial. *The Lancet, 387*(10013), 53–60. https://doi.org/10.1016/S0140-6736(15)00056-2

McHugh, M. L. (2012). Interrater reliability: The kappa statistic. *Biochemia Medica, 22*(3), 276–282.

McLaughlin, M. L., Hou, J., Meng, J., Hu, C.-W., An, Z., Park, M., & Nam, Y. (2016). Propagation of information about preexposure prophylaxis (PrEP) for HIV prevention through Twitter. *Health Communication, 31*(8), 998–1007. https://doi.org/10.1080/10410236.2015.1027033

Meyers, K., Wu, Y., Shin, K.-Y., Hou, J., Hu, Q., Duan, J., Li, Y., & He, X. (2021). Salient constructs for the development of shared decision-making tools for HIV pre-exposure prophylaxis uptake and regimen choice: Behaviors, behavioral skills, and beliefs. *AIDS Patient Care and STDs, 35*(6), 195–203. https://doi.org/10.1089/apc.2021.0066

Molina, J.-M., Capitant, C., Spire, B., Pialoux, G., Cotte, L., Charreau, I., Tremblay, C., Le Gall, J.-M., Cua, E., Pasquet, A., Raffi, F., Pintado, C., Chidiac, C., Chas, J., Charbonneau, P., Delaugerre, C., Suzan-Monti, M., Loze, B., Fonsart, J., … Delfraissy, J.-F. (2015). On-demand preexposure prophylaxis in men at high risk for HIV-1 infection. *New England Journal of Medicine, 373*(23), 2237–2246. https://doi.org/10.1056/NEJMoa1506273

Molina, J.-M., Charreau, I., Spire, B., Cotte, L., Chas, J., Capitant, C., Tremblay, C., Rojas-

Castro, D., Cua, E., Pasquet, A., Bernaud, C., Pintado, C., Delaugerre, C., Sagaon-Teyssier, L., Mestre, S. L., Chidiac, C., Pialoux, G., Ponscarme, D., Fonsart, J., … Rabian, C. (2017). Efficacy, safety, and effect on sexual behaviour of on-demand pre-exposure prophylaxis for HIV in men who have sex with men: An observational cohort study. *The Lancet HIV, 4*(9), e402–e410. https://doi.org/10.1016/S2352-3018(17)30089-9

Patel, V. V., Masyukova, M., Sutton, D., & Horvath, K. J. (2016). Social media use and HIV-related risk behaviors in young Black and Latino gay and bi men and transgender individuals in New York City: Implications for online interventions. *Journal of Urban Health : Bulletin of the New York Academy of Medicine, 93*(2), 388–399. https://doi.org/10.1007/s11524-016-0025-1

Pennebaker, J. W., Booth, R. J., Boyd, R. L., & Francis, M. E. (2015). Linguistic Inquiry and Word Count: LIWC2015. *Pennebaker Conglomerates.* www.LIWC.net

Pornpitakpan, C. (2004). The persuasiveness of source credibility: A critical review of five decades' evidence. *Journal of Applied Social Psychology, 34*(2), 243–281. https://doi.org/10.1111/j.1559-1816.2004.tb02547.x

Rains, S. A., Leroy, G., Warner, E. L., & Harber, P. (2021). Psycholinguistic markers of COVID-19 conspiracy tweets and predictors of tweet dissemination. *Health Communication, 0*(0), 1–10. https://doi.org/10.1080/10410236.2021.1929691

Roth, A., Felsher, M., Tran, N., Bellamy, S., Martinez-Donate, A., Krakower, D., Szep, Z., Roth, A., Felsher, M., Tran, N., Bellamy, S., Martinez-Donate, A., Krakower, D., & Szep, Z. (2019). Drawing from the Theory of Planned Behaviour to examine pre-exposure prophylaxis uptake intentions among heterosexuals in high HIV prevalence neighbourhoods in Philadelphia, Pennsylvania, USA: An

observational study. *Sexual Health, 16*(3), 218–224. https://doi.org/10.1071/SH18081

Sayyadiharikandeh, M., Varol, O., Yang, K.-C., Flammini, A., & Menczer, F. (2020). Detection of novel social bots by ensembles of specialized classifiers. In *Proceedings of the 29th ACM International Conference on Information & Knowledge Management* (pp. 2725–2732). https://doi.org/10.1145/3340531.3412698

Schwartz, J., & Grimm, J. (2017). PrEP on Twitter: Information, barriers, and stigma. *Health Communication, 32*(4), 509–516. https://doi.org/10.1080/10410236.2016.1140271

Surian, D., Nguyen, D. Q., Kennedy, G., Johnson, M., Coiera, E., & Dunn, A. G. (2016). Characterizing Twitter discussions about HPV vaccines using topic modeling and community detection. *Journal of Medical Internet Research, 18*(8), e6045. https://doi.org/10.2196/jmir.6045

Tausczik, Y. R., & Pennebaker, J. W. (2010). The psychological meaning of words: LIWC and computerized text analysis methods. *Journal of Language and Social Psychology, 29*(1), 24–54. https://doi.org/10.1177/0261927X09351676

Thigpen, M. C., Kebaabetswe, P. M., Paxton, L. A., Smith, D. K., Rose, C. E., Segolodi, T. M., Henderson, F. L., Pathak, S. R., Soud, F. A., Chillag, K. L., Mutanhaurwa, R., Chirwa, L. I., Kasonde, M., Abebe, D., Buliva, E., Gvetadze, R. J., Johnson, S., Sukalac, T., Thomas, V. T., … Brooks, J. T. (2012). Antiretroviral preexposure prophylaxis for heterosexual HIV transmission in Botswana. *New England Journal of Medicine, 367*(5), 423–434. https://doi.org/10.1056/NEJMoa1110711

Tran, N. K., Felsher, M., Pol, B. V. D., Bellamy, S. L., McKnight, J., & Roth, A. M. (2021). Intention to initiate and uptake of PrEP among women who injects drugs in a demonstration project: An application of the theory of planned behavior. *AIDS*

Care, 33(6), 746–753. https://doi.org/10.1080/09540121.2021.1874267

U.S. Department of Health and Human Services. (2021). HIV National Strategic Plan: A Roadmap to End the Epidemic for the United States 2021–2025. https://hivgov-prod-v3.s3.amazonaws.com/s3fs-public/HIV-National-Strategic-Plan-2021-2025.pdf

U.S. Food and Drug Administration. (2019). FDA approves second drug to prevent HIV infection as part of ongoing efforts to end the HIV epidemic. https://www.fda.gov/news-events/press-announcements/fda-approves-second-drug-prevent-hiv-infection-part-ongoing-efforts-end-hiv-epidemic

Walsh-Buhi, E., Houghton, R. F., Lange, C., Hockensmith, R., Ferrand, J., & Martinez, L. (2021). Pre-exposure prophylaxis (PrEP) information on Instagram: Content analysis. *JMIR Public Health and Surveillance, 7*(7), e23876. https://doi.org/10.2196/23876

Wang, Z., Mo, P. K. H., Ip, M., Fang, Y., & Lau, J. T. F. (2020). Uptake and willingness to use PrEP among Chinese gay, bisexual and other men who have sex with men with experience of sexualized drug use in the past year. *BMC Infectious Diseases, 20*(1), 299. https://doi.org/10.1186/s12879-020-05024-4

Wojcik, S., & Hughes, A. (2019, April 24). How Twitter users compare to the general public. *Pew Research Center.* https://www.pewresearch.org/internet/2019/04/24/sizing-up-twitter-users/

Zacharias, C. (2020). twint: An advanced Twitter scraping & OSINT tool. (2.1.20) [Python]. https://github.com/twintproject/twint

Zhang, J., Le, G., Larochelle, D., Pasick, R., Sawaya, G. F., Sarkar, U., & Centola, D. (2019). Facts or stories? How to use social media for cervical cancer prevention: A

multi-method study of the effects of sender type and content type on increased message sharing. *Preventive Medicine, 126*, 105751. https://doi.org/10.1016/j.ypmed.2019.105751

Zhang, J., Xue, H., Calabrese, C., Chen, H., & Dang, J. H. T. (2021). Understanding human papillomavirus vaccine promotions and hesitancy in Northern California through examining public Facebook pages and groups. *Frontiers in Digital Health, 3*. https://doi.org/10.3389/fdgth.2021.683090

Supplementary Material

Nonparametric Tests

Differences in Coded Tweets After the Approval of Descovy

Differences in Coded Tweets between Source Type

Differences in Emotions and Sentiments after the Approval of Descovy

Differences in Emotions and Sentiments by Source Type

Removing "Bots"

Table S1. Characteristics of PrEP Tweets by FDA Approval of Descovy

Table S2. Types of Actual Barriers by FDA Approval of Descovy

Table S3. Characteristics of PrEP Tweets by Source Type

Table S4. Emotions and Sentiments of PrEP Tweets by FDA Approval

Table S5. Emotions and Sentiments of PrEP Tweets by Source Type

Nonparametric Tests

Differences in Coded Tweets After the Approval of Descovy

Among coded tweets (N = 1008), Mann-Whitney U tests were conducted to compare differences in the Theory of Planned Behavior (TPB) constructs (Fishbein & Ajzen, 2011) and source before and after the approval of Descovy. These nonparametric tests were conducted to further ensure the robustness of these results.

TPB Constructs. There were significantly more information tweets about PrEP after the approval of Descovy than before, $U = 116193.5$, $p = .003$. Further, there were significantly less tweets discussing barriers to PrEP after the approval of Descovy, $U = 106727$, $p < .001$. There were no significant differences in attitudes ($p = .247$), norms ($p = .113$), perceived behavioral control ($p = .157$), and intentions/behavior ($p = .111$) after the approval of Descovy.

Specific Barriers. Among tweets discussing barriers to PrEP, there were significantly less tweets discussing costs after the approval of Descovy than before, $U = 121057$, $p = .040$. Further, there were significantly less tweets discussing access after the approval of Descovy than before, $U = 114241$, $p < .001$. There were no differences in issues with pharmaceutical company ($p = .237$) and stigma ($p = .393$).

Source Type. There were no differences in source type (individual versus organization) before and after the approval of Descovy ($p = .692$).

Differences in Coded Tweets between Source Type

Among coded tweets ($N = 1008$), Mann-Whitney U tests were conducted to compare differences in TPB constructs by source type (individual versus organization). These nonparametric tests further ensure the robustness of these results.

TPB Constructs. There were significantly more informational tweets posted by organizations than individuals, $U = 71445.5$, $p < .001$. In addition, there were significantly more tweets expressing attitudes posted by individuals than organizations, $U = 88168.5$, $p < .001$. Further, there were significantly more tweets expressing intentions/behavior posted by individuals than organizations, $U = 87403.5$, $p < .001$. There were no differences between source type for norms ($p = .994$), perceived behavioral control ($p = .065$), and overall barriers ($p = .422$).

Specific Barriers. Among barrier tweets, there were significantly more individual tweets about costs ($U = 12439.5$, $p = .013$) and stigma ($U = 13503.5$, $p = .016$) compared to organizational tweets. However, there were significantly more organizational tweets that discuss access compared to individual tweets, $U = 11563$, $p < .001$. There were no differences regarding issues with pharmaceutical company by source type ($p = .938$).

Differences in Emotions and Sentiments after the Approval of Descovy

For the full dataset ($N = 16139$), Mann-Whitney U tests were conducted to compare differences in positive emotion, negative emotion, anxiety, anger, and sadness before and after the approval of Descovy. These nonparametric

tests further ensure the robustness of these results.

Positive and Negative Emotion. There were significantly more tweets that expressed positive emotion after the approval of Descovy than before, U = 30240973, p < .001. However, there were no significant differences in negative emotion (p = .576).

Discrete Emotion. There were significantly less tweets that expressed anger after the approval of Descovy than before, U = 31923004.5, p < .001. However, there were significantly more tweets that expressed sadness after the approval of Descovy than before, U = 32293740, p = .041. There were no significant differences in anxiety (p = .543).

Differences in Emotions and Sentiments by Source Type

Mann-Whitney U tests were conducted to compare differences in positive emotion, negative emotion, anxiety, anger, and sadness by source type (individual vs. organization). These nonparametric tests further ensure the robustness of these results.

Positive and Negative Emotion. Tweets posted by individuals expressed significantly more positive emotion than organizational tweets, U = 86531, p = .014. However, there were no significant differences in negative emotion (p = .638).

Discrete Emotion. Organizational tweets expressed significantly more anxiety compared to individual tweets, U = 90177, p = .018. However, there were no significant differences in anger (p = .159) or sadness (p = .832).

Removing "Bots"

In the following analyses, we ran Botometer (Sayyadiharikandeh et al., 2020), a bot classification tool based on a machine learning algorithm that outputs a score indicating whether a tweet was made from a human or a bot. Complete Automation Probability (CAP) scores provide the probability (from 0 to 1) that an account with a X% score or greater is likely a bot. Scores closer to 0 indicate that the tweet was likely from a human, and scores closer

to 1 indicate that the tweet is likely from a bot account. We utilized a cutoff CAP score of 80% and re-ran analysis to examine any discrepancies between the "no bot" sample and the full coded sample. It is important to note that though this machine learning tool has been trained and extracts over 1000 different features to determine bot score (*Botometer*, 2021), these tools are not perfect and may incorrectly classify some tweets. For example, organizations that post content systematically may be identified as a bot by the classifier.

Tables S1-4 display the re-run analyses with only the "no bot" sample. Results remain consistent among all but one finding. In the "no bot" sample, there was not a significant difference in access tweets before and after the FDA approval; however, these findings should be taken with caution. This may be likely due the smaller sample size of barrier tweets or potentially by an error in the bot classification of the tweets.

Table S1. Characteristics of PrEP Tweets by FDA Approval of Descovy (N = 786)

Variable	Before FDA Approval N (%)	After FDA Approval N (%)	p-value
TPB Construct			
Attitude	26 (6.5%)	27 (7.0%)	.797
Perceived Norm	8 (2.0%)	3 (.7%)	.142
Perceived behavioral control	5 (1.3%)	8 (2.1%)	.371
Intention/Behavior	25 (6.3%)	37 (9.6%)	.087
Information/ Knowledge	82 (20.6%)	107 (27.6%)	.020
Actual barriers	176 (44.1%)	122 (31.5%)	<.001
Other	77 (19.3%)	83 (21.4%)	.454
Source			
Individual	323 (81.0%)	302 (78.0%)	.311
Organization	76 (19.0%)	85 (22.0%)	

Notes: P-values were calculated from χ^2 tests; all cells had expected values over 5.

Table S2. Types of Actual Barriers by FDA Approval of Descovy (N = 298)

Variable	Before FDA Approval N (%)	After FDA Approval N (%)	p-value
Cost	68 (38.6%)	50 (41.0%)	.684
Access	69 (39.2%)	37 (30.3%)	.115
Issues with Pharmaceutical Company	33 (18.8%)	27 (22.1%)	.474
Stigma	6 (3.4%)	8 (6.6%)	.207

Notes: P-values were calculated from χ^2 tests; all cells had expected values over 5.

Table S3. Characteristics of PrEP Tweets by Source (N = 786)

Variable	Organization	Individual	p-value
TPB Construct			
Attitude	0 (0%)	53 (8.4%)	<.001
Perceived norm	3 (1.9%)	8 (1.3%)	.705 [a]
Perceived behavioral control	1 (.6%)	12 (1.9%)	.485 [a]
Intention/behavior	1 (.6%)	61 (9.8%)	<.001
Information/ knowledge	73 (45.3%)	116 (18.6%)	<.001
Actual barriers	67 (41.6%)	231 (37.0%)	.278
Other	16 (9.9%)	144 (23.0%)	<.001

Notes: P-values were calculated from χ^2 tests; all cells had expected values over 5.

[a] P-values were calculated using Fisher's exact tests.

Table S4. Emotions and Sentiments of PrEP Tweets by FDA Approval (N = 12618)

Variable	t-statistic	df	p-value
Positive emotion	-5.13	12575.78	<.001
Negative emotion	1.32	12597.52	.187
Anxiety	.57	12614.66	.571
Anger	3.84	12132.18	<.001
Sadness	.40	12485.72	.689

Notes: T-statistics were conducted using Welch's t-tests for unequal variances.

Table S5. Emotions and Sentiments of PrEP Tweets by Source Type (N = 786)

Variable	t-statistic	df	p-value
Positive emotion	-2.83	357.24	.005
Negative emotion	-.48	298.50	.635
Anxiety	2.07	226.78	.039
Anger	-.471	305.89	.638
Sadness	-1.05	362.16	.293

Notes: T-statistics were conducted using Welch's t-tests for unequal variances.

References

Botometer. (2021). Botometer. https://botometer.osome.iu.edu/

Fishbein, M., & Ajzen, I. (2011). *Predicting and changing behavior: The reasoned action approach*. Psychology Press. https://doi.org/10.4324/9780203838020

Sayyadiharikandeh, M., Varol, O., Yang, K.-C., Flammini, A., & Menczer, F. (2020). Detection of novel social bots by ensembles of specialized classifiers. *Proceedings of the 29th ACM International Conference on Information & Knowledge Management*, 2725–2732. https://doi.org/10.1145/3340531.3412698